

CY1010 – Environmental Chemistry

Practice Question Paper 2 (50 marks)

Compiled from course slides — not an official department paper

Student name: _____

Roll No: _____

Department: _____

Each question carries 1 mark. Choose the single best answer and write the corresponding letter on the "Ans" line. This paper does not repeat questions from Practice Paper 1.

1. According to the slides, the nitrogen cycle proceeds via nitrogen fixation, ammonification, nitrification, assimilation, decomposition, and

- a) Photosynthesis
- b) Denitrification
- c) Combustion
- d) Sublimation

Ans: _____

2. The four steps of fossil fuel formation described in the slides are photosynthesis exceeding respiration, detritus accumulation, burial of detritus, and

- a) Fractional distillation
- b) Pressure and heat (metamorphosis)
- c) Bacterial fermentation
- d) Catalytic cracking

Ans: _____

3. According to the slides, fossil fuel reserves are estimated to last approximately

- a) 50 years for oil & natural gas; 380 years for coal
- b) 100 years for oil & gas; 100 years for coal
- c) 380 years for oil & gas; 50 years for coal
- d) 10 years for oil & gas; 500 years for coal

Ans: _____

4. Crude oil is stated in the slides to supply approximately what fraction of the world's energy?

- a) One-tenth
- b) One-third
- c) Two-thirds
- d) Nine-tenths

Ans: _____

5. In a hydrogen fuel cell, the cathode (with a platinum catalyst) is responsible for

- a) Splitting hydrogen molecules into protons and electrons
- b) Oxygen reduction by reaction with protons, producing water
- c) Storing electrical energy chemically
- d) Compressing hydrogen gas

Ans: _____

6. The surface temperature of the Sun, as given in the slides, is approximately

- a) 5500 °C
- b) 15,000 °C
- c) 1 million °C
- d) 100 °C

Ans: _____

7. The total power output of the Sun is given in the slides as approximately

- a) $3.84 \times 10^{16} \text{ W}$
- b) $3.84 \times 10^{20} \text{ W}$
- c) $3.84 \times 10^{26} \text{ W}$
- d) $3.84 \times 10^{30} \text{ W}$

Ans: _____

8. The intensity of solar radiation at Earth's orbital distance is calculated in the slides to be approximately

- a) 137.7 W/m^2
- b) 1377 W/m^2
- c) $13,770 \text{ W/m}^2$
- d) 1.377 W/m^2

Ans: _____

9. Which factors are identified in the slides as the main causes of wind?

- a) Uneven solar heating, the Coriolis effect, and local geographical conditions
- b) Ocean currents alone
- c) Volcanic activity
- d) Lunar gravitational pull alone

Ans: _____

10. According to the slides, in 2025 wind energy supplied roughly what share of world electricity?

- a) 1%
- b) 8%
- c) 25%
- d) 50%

Ans: _____

11. Pumped hydro is described in the slides as contributing what share of large-scale, long-duration stationary energy storage?

- a) 50%
- b) 75%
- c) 98%
- d) 10%

Ans: _____

12. Which of the following is NOT among the strategies to save energy listed in the slides?

- a) Greater use of mass transportation
- b) Greater use of energy-efficient appliances
- c) Increasing single-use packaging
- d) Recycling

Ans: _____

13. Geothermal steam wells described in the slides can be bored to depths of up to 1000 m, releasing steam and hot water at temperatures of about

- a) 50–80 °C
- b) 100–150 °C
- c) 200–300 °C
- d) 500–600 °C

Ans: _____

14. Ocean Thermal Energy Conversion (OTEC) exploits a temperature gradient of about how many degrees between the ocean surface and depth, according to the slides?

- a) 2–5 °C
- b) 20–25 °C

- c) 50–60 °C
 - d) 100 °C
- Ans: _____

15. In tidal power generation, the period when sea water is above the mean sea level is called

- a) Ebb tide
- b) Flood tide
- c) Neap tide
- d) Spring surge

Ans: _____

16. The Sihwa Lake Tidal Power Station, described in the slides as the world's largest tidal power installation (254 MW), is located in

- a) France
- b) South Korea
- c) India
- d) Iceland

Ans: _____

17. Non-edible biomass, as defined in the slides, includes

- a) Cereals, starch, and sugars
- b) Agricultural waste residues (lignocellulosic biomass), forest residues, and grassy materials
- c) Refined vegetable oils
- d) Dairy products

Ans: _____

18. Which statement correctly distinguishes starch from cellulose, as described in the slides?

- a) Starch is a linear/branched glucose polymer with intramolecular H-bonds (edible); cellulose is a linear glucose polymer with intra- and intermolecular H-bonds (non-edible)
- b) Starch is non-edible and cellulose is edible
- c) Both are chemically identical in digestibility
- d) Starch contains no glucose units

Ans: _____

19. According to the slides, catalysts increase reaction rates mainly by

- a) Permanently raising the reaction temperature
- b) Lowering the overall activation energy of the reaction
- c) Consuming the reactants faster
- d) Raising the enthalpy of the products

Ans: _____

20. The enzyme nitrogenase, found in bacteria living in root nodules of legumes such as clover and alfalfa, catalyzes the conversion of

- a) NH_3 to N_2
- b) N_2 to NH_3
- c) NO_3^- to N_2O
- d) NH_3 to NO_3^-

Ans: _____

21. In a catalytic converter, NO and NO_2 are converted (with CO and unburned hydrocarbons oxidised separately) into

- a) N_2 and O_3
- b) NH_3
- c) N_2 and O_2
- d) N_2O

Ans: _____

22. According to the slides, the scenario of rising global carbon emissions is driven mainly by population growth, industrialization, and

- a) Reduced fossil fuel use
- b) Heavy dependence on fossil fuels
- c) Ocean acidification
- d) Declining energy demand

Ans: _____

23. According to the slides' estimate, approximately what fraction of the mass of Earth's atmosphere lies within the troposphere?

- a) 20%
- b) 50%
- c) 80%
- d) 99%

Ans: _____

24. Based on the slides, roughly what fraction of global energy consumption (~500 EJ/year) is obtained by burning fossil fuels?

- a) 10–15%
- b) 40–50%
- c) 80–85%
- d) Nearly 100%

Ans: _____

25. According to the slides' breakdown of crude oil composition by weight, carbon accounts for approximately

- a) 40–50%
- b) 83–85%
- c) 95–99%
- d) 10–14%

Ans: _____

26. Chlorofluorocarbons (CFCs) are characterized in the slides mainly by

- a) Low volatility and high boiling points
- b) High volatility, boiling points close to 0 °C, and the ability to break down stratospheric ozone
- c) Their inability to circulate in refrigeration systems
- d) High solubility in water

Ans: _____

27. Unlike CFCs, hydrofluorocarbons (HFCs) do not damage the ozone layer because they

- a) Do not absorb infrared radiation at all
- b) Contain no chlorine
- c) Are never used in refrigeration
- d) Break down instantly in sunlight

Ans: _____

28. In the GWP table given in the slides, which gas has an atmospheric lifetime of about 12.2 years and a GWP of 25 relative to CO₂?

- a) Nitrous oxide
- b) Methane
- c) Sulphur hexafluoride
- d) A representative perfluorocarbon

Ans: _____

29. According to the slides, direct air capture (DAC) uses two broad types of capturing agents:

- a) Liquid solvents and solid adsorbents

- b) Metals and ceramics
 - c) Enzymes and polymers
 - d) Acids and salts
- Ans: _____

30. Amine-functionalized solid sorbents are noted in the slides for allowing CO₂ desorption/regeneration at relatively low temperatures of about

- a) 900 °C
 - b) 500–600 °C
 - c) 50–120 °C
 - d) 0 °C
- Ans: _____

31. According to the slides, metal–organic frameworks (MOFs) can have empty space within their framework of up to

- a) 20%
 - b) 50%
 - c) 94%
 - d) 10%
- Ans: _____

32. Specific surface area, one of the porosity parameters defined in the slides, is calculated as

- a) Mass of the solid divided by total surface area
 - b) Total surface area divided by mass of the solid
 - c) Volume of pores divided by volume of solid
 - d) Total pore volume divided by total surface area
- Ans: _____

33. Polycarbonates synthesized from CO₂ via ring-opening copolymerization of epoxides commonly use which catalyst, according to the slides?

- a) ZnEt₂
 - b) NaOH
 - c) Pt
 - d) Fe
- Ans: _____

34. According to the global carbon budget slide, if current trends continue, global temperature rise by the year 2100 is projected to exceed

- a) 1 °C
 - b) 2 °C
 - c) 4 °C
 - d) 10 °C
- Ans: _____

35. Direct Air Capture (DAC) is highlighted in the slides as significant because it is described as

- a) The cheapest carbon capture method available
 - b) The only negative emission technology
 - c) Unnecessary given renewable energy growth
 - d) Usable only above 1000 ppm atmospheric CO₂
- Ans: _____

36. The UN's 17 Sustainable Development Goals are described in the slides as

- a) A shared blueprint for peace and prosperity for people and the planet
- b) A binding legal treaty only for developed nations
- c) A set of chemical safety regulations
- d) A framework exclusive to climate finance

Ans: _____

37. Wangari Maathai, awarded the 2004 Nobel Peace Prize, founded which movement described in the slides?

- a) Chipko movement
- b) Green Belt Movement
- c) Silent Valley movement
- d) Narmada Bachao Andolan

Ans: _____

38. Which Indian environmentalist is referred to in the slides as the "Waterman of India"?

- a) Salim Ali
- b) M. C. Mehta
- c) Rajender Singh
- d) Sunderlal Bahuguna

Ans: _____

39. Salim Ali, honoured with the Padma Vibhushan in 1976, is described in the slides as

- a) The "Birdman of India" for his work in ornithology
- b) The founder of the Chipko movement
- c) The "Waterman of India"
- d) A pioneer of green hydrogen technology

Ans: _____

40. According to the slides, causes of oil spills include tanker accidents, offshore drilling accidents, pipeline leaks, equipment failure, and

- a) Overfishing
- b) Natural disasters
- c) Excess biodiesel production
- d) Ozone depletion

Ans: _____

41. Which oil-spill remediation method is described in the slides as achieving high cleanup efficiency but producing smoke and harmful air pollutants?

- a) Mechanical remediation
- b) In-situ burning
- c) Biological remediation
- d) Chemical remediation

Ans: _____

42. Biological remediation of oil spills is highlighted in the slides as

- a) The most expensive and slowest method available
- b) An environmentally friendly and cost-effective cleanup method using microorganisms
- c) Ineffective in marine environments
- d) Requiring high-temperature combustion

Ans: _____

43. According to the slides, biodegradation of spilled oil occurs at the oil–water interface in the presence of

- a) Oxygen
- b) Only in the absence of oxygen
- c) Ultraviolet light exclusively
- d) High salinity alone

Ans: _____

44. Among the enzymes involved in microbial degradation of organic pollutants, which is described in the slides as occurring mainly in fungi?

- a) Dehydrogenases
- b) Hydrolases
- c) Laccases
- d) General peroxidases

Ans: _____

45. Oxidation of naphthalene with KMnO_4 under acidic conditions, as shown in the slides, produces

- a) Phthalic acid
- b) 1,4-Naphthoquinone
- c) Phthalic anhydride
- d) Phthalaldehyde

Ans: _____

46. According to the slides, the first metabolite formed during methanol oxidation in the body, responsible for neuronal damage, is

- a) Formic acid
- b) Formaldehyde
- c) Acetaldehyde
- d) Methyl ester

Ans: _____

47. According to the slides, adding starch granules (5–20%) to polyethylene is used to manufacture shopping bags that are claimed to be

- a) Fire-resistant
- b) Biodegradable
- c) Waterproof
- d) Electrically conductive

Ans: _____

48. Which of the following is listed in the slides as one of the categories of organic chemicals found in the environment?

- a) Noble gases
- b) Pharmaceutical agents (medicines, antibiotics, hormones)
- c) Radioactive isotopes only
- d) Metals and alloys

Ans: _____

49. In the slides' example of the epoxidation of styrene (assuming 100% yield, with m-chlorobenzoic acid as a co-product), the atom economy of the reaction is given as

- a) 100%
- b) 77%
- c) 50%
- d) 23%

Ans: _____

50. According to the case study in the slides, an alternative process for producing ethylene oxide that uses molecular oxygen instead of chlorine is reported to

- a) Generate the same amount of waste as the conventional process
- b) Generate more than 16 times less waste, and eliminate wastewater formation
- c) Generate twice as much waste
- d) Eliminate CO_2 emissions but keep the same waste level

Ans: _____

Answer Key

1.	b	26.	b
2.	b	27.	b
3.	a	28.	b
4.	b	29.	a
5.	b	30.	c
6.	a	31.	c
7.	c	32.	b
8.	b	33.	a
9.	a	34.	c
10.	b	35.	b
11.	c	36.	a
12.	c	37.	b
13.	c	38.	c
14.	b	39.	a
15.	b	40.	b
16.	b	41.	b
17.	b	42.	b
18.	a	43.	a
19.	b	44.	c
20.	b	45.	a
21.	c	46.	b
22.	b	47.	b
23.	c	48.	b
24.	c	49.	d
25.	b	50.	b